Disentangling Demand and Supply Shocks in the Crude Oil Market

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Reference paper: Kilian, Lutz. 2009. "Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market." American Economic Review 99 (3): 1053–69.

Motivation

$$e_t = \begin{pmatrix} e_t^{\Delta prod} \\ e_t^{rea} \\ e_t^{rpo} \end{pmatrix} = \begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} \epsilon_t^{\text{oil supply}} \\ \epsilon_t^{\text{aggregate demand}} \\ \epsilon_t^{\text{oil-specific demand}} \end{pmatrix}$$

Two problem about previous models:

- Reverse causality
- Not accounting for different types of shocks

Using a Bayesian SVAR model inspired by Kilian (2009) to measure:

1. Effects of different shocks on macroeconomic performance and oil prices
2. Causal inference on oil price shocks to US macroeconomic aggregates

Three types of shocks:

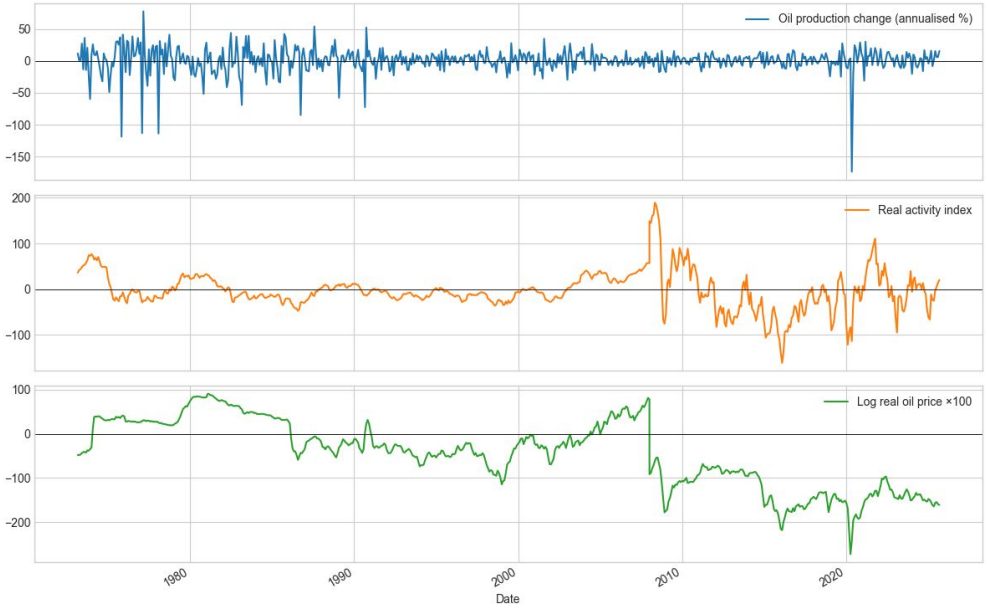
1. Oil supply shocks
2. Aggregate demand shocks (Global industrial commodity)
3. Precautionary oil demand shocks

Data

- **d_prod**: Crude Oil production log-difference (from EIA)
- **rea**: “Kilian index” - Dry cargo single voyage ocean freight rates
- **rpo**: real oil price - imported refiner acquisition cost of crude oil divided be CPI, not seasonally adjusted

Range: 1973 January - 2025 September (**Monthly**)

* Kilian index: construct from monthly report of “Shipping Statistics and Economics” by Drewry Shipping Consultants Ltd; Can be publicly download via Federal reserve bank of Dallas



Model & Method

- NIW - BSVAR(24) model with short-run restriction
- Gibbs Sampler with 12000 draws, 2000 burn-in
- Bootstrap for CI: 20000; Block: 4 (Quarterly)

$$A_0 z_t = a + \sum_{i=1}^{p} A_i z_{i-1} + \epsilon_t$$

Where $z_t = (\Delta prod_t, rea_t, rpo_t)'$ and $e_t = A_0^{-1}\epsilon_t$ can be decomposed as:

$$e_t = \begin{pmatrix} e_t^{\Delta prod} \\ e_t^{rea} \\ e_t^{rpo} \end{pmatrix} = \begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} \epsilon_t^{\text{oil supply}} \\ \epsilon_t^{\text{aggregate demand}} \\ \epsilon_t^{\text{oil-specific demand}} \end{pmatrix}$$

We estimate a Bayesian VAR of the form:

$$z_t = c + \sum_{i=1}^{p} A_i z_{t-i} + e_t$$

where $e_t \sim N(0, \Sigma)$. The model is estimated using a Normal-Inverse Wishart prior:

$$\vec{B} \mid \Sigma \sim \mathcal{N}(\vec{B_0}, \Sigma \otimes V_0), \quad \Sigma \sim \mathcal{IW}(S_0, \nu_0)$$

where $\vec{B} = (c, A_1, ..., A_p)'$. Posterior inference is conducted via a Gibbs sampler. Structural shocks are identified by imposing a recursive structure, consistent with Kilian (2009), using a Cholensky decomposition:

$$e_t = A_0^{-1}\epsilon_t, \quad \epsilon_t \sim N(0, I)$$

With covariance matrix:

$$\Sigma = A_0^{-1}(A_0^{-1})'$$

Gibbs Sampler algorithm

We use Gibbs sampler to draw sample for the Bayesian model. The algorithm is as follow:
Initialize Σ (e.g. from OLS residuals). For $s = 1, 2, \ldots, S$:

1. Draw $B^{(s)} \sim \mathcal{MN}(B_n^{(s-1)}, V_n, \Sigma^{(s-1)})$ with $B_n^{(s-1)}, V_n$ computed from X, Y and the prior as above.
2. Draw $\Sigma^{(s)} \sim \mathcal{IW}(\nu_n, S_n)$ using $B^{(s)}$ in E and S_n.

Discard a fair amount of initial burn-in of draws (2000 used here); retain the rest for posterior summaries of B and Σ (10000 remaining).

Then, we generate the IRFs and FEVD with comparison with the original result.

Regression model for US Economy Impact

The quarterly shocks are constructed by averaging the monthly structural innovations:

$$\hat{\zeta}_{jt} = \frac{1}{3}\sum_{i=1}^{3} \hat{\varepsilon}_{j,t,i}, \quad j = 1, 2, 3 \tag{1}$$

where $\hat{\varepsilon}_{j,t,i}$ refers to the estimated residual for the jth structural shock in the ith month of the tth quarter of the sample.

The effects on US macroeconomic aggregates are examined based on the regressions:

$$\Delta y_t = \alpha_j + \sum_{i=0}^{12} \phi_{ji}\hat{\zeta}_{jt-i} + u_{jt}, \quad j = 1, 2, 3 \tag{2}$$

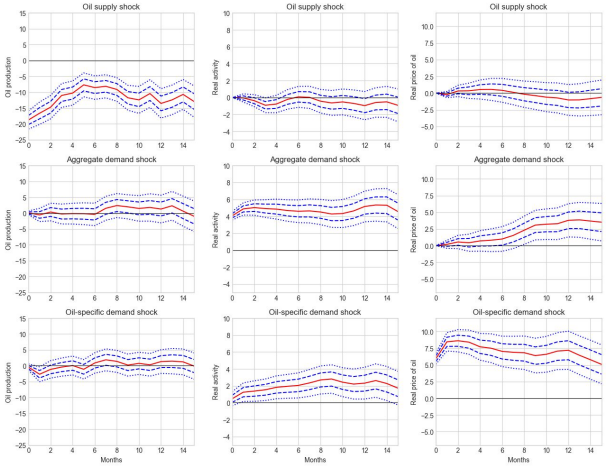
and

$$\pi_t = \delta_j + \sum_{i=0}^{12} \psi_{ji}\hat{\zeta}_{jt-i} + v_{jt}, \quad j = 1, 2, 3 \tag{3}$$

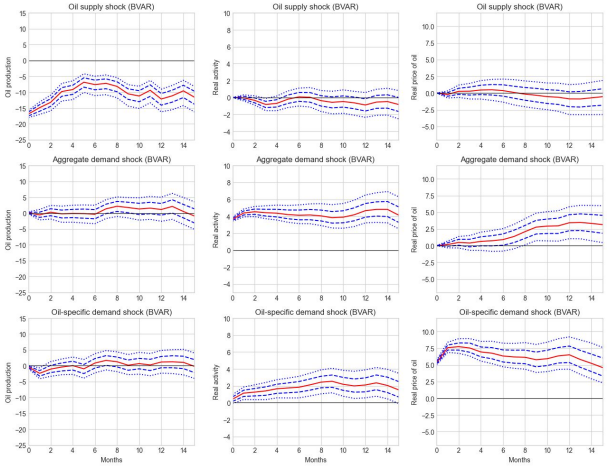
where u_{jt} and v_{jt} are potentially serially correlated errors.

Empirical result - Replication

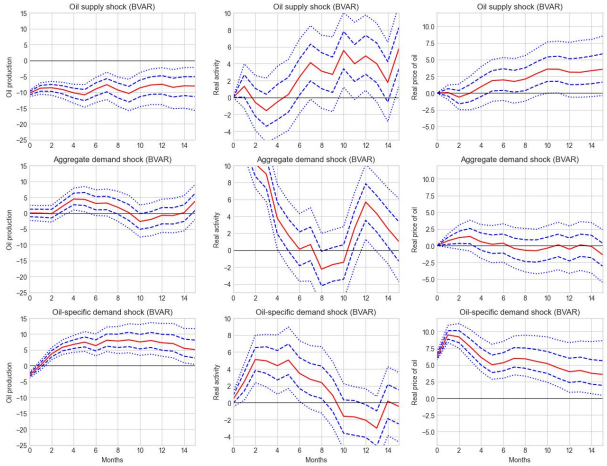
SVAR



BSVAR

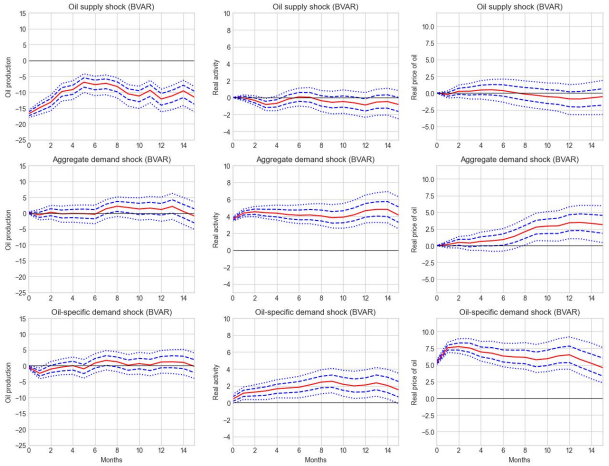


BSVAR (Out-of-sample)

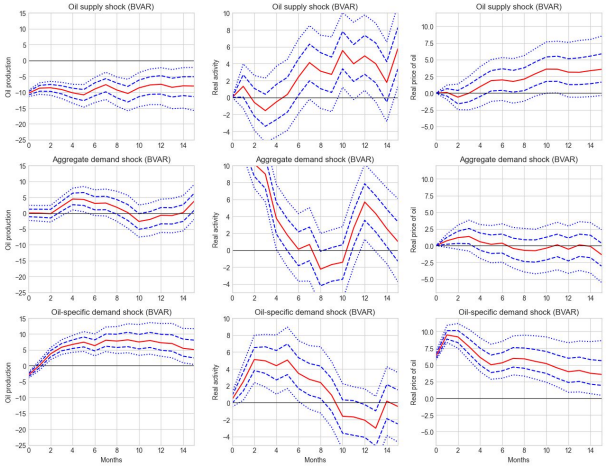


Empirical result - IRF

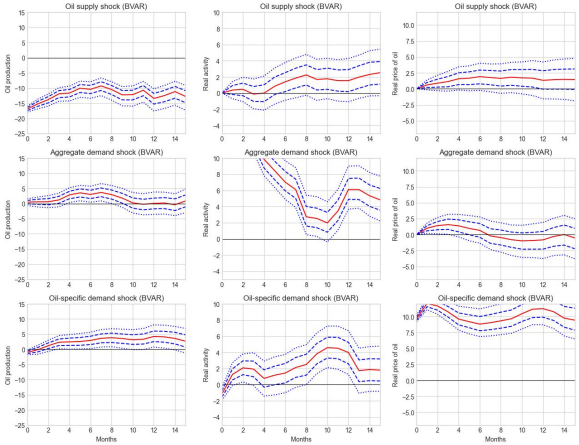
BSVAR



BSVAR(Out-of-sample)

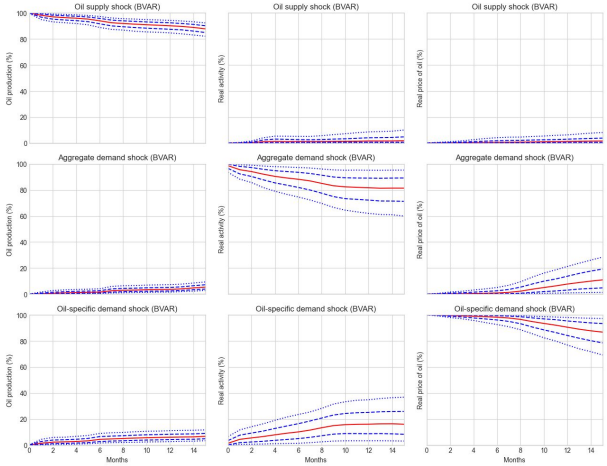


BSVAR (Full)



Empirical result - FEVD

BSVAR



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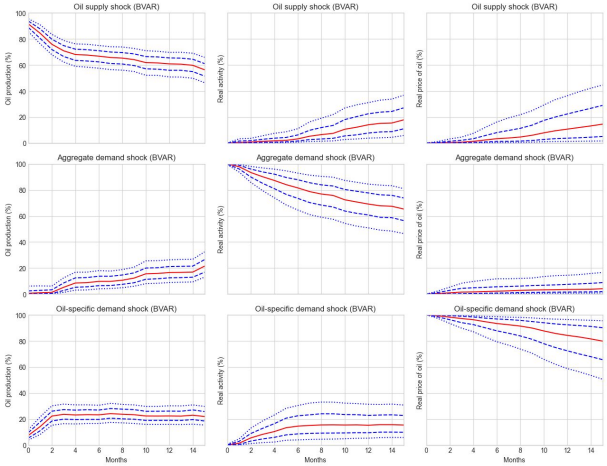
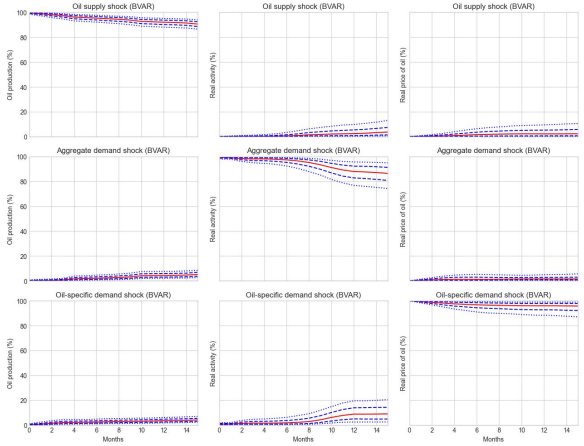
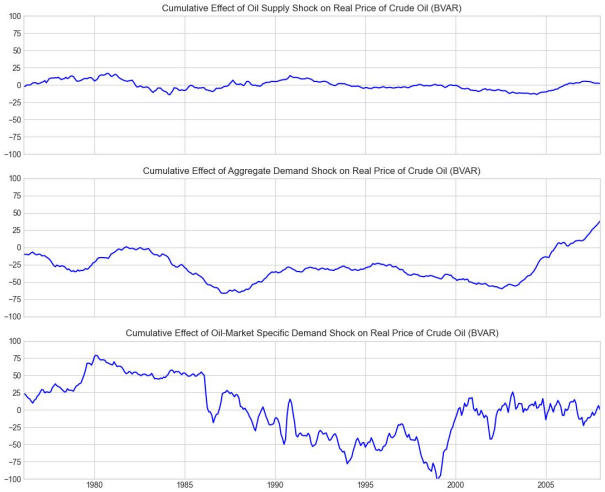


BSVAR (Full)

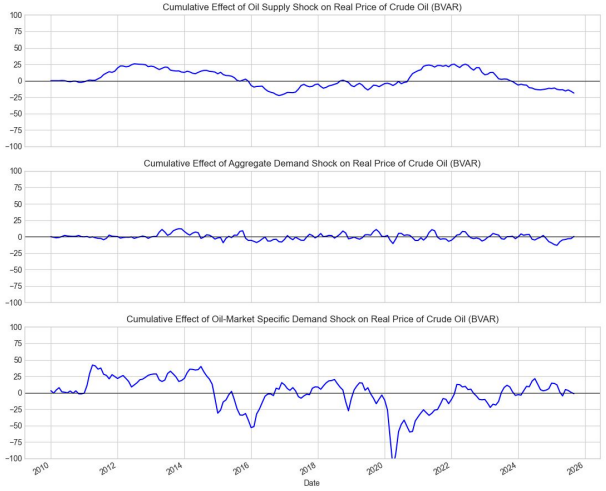


Empirical result - Cumulative Effect

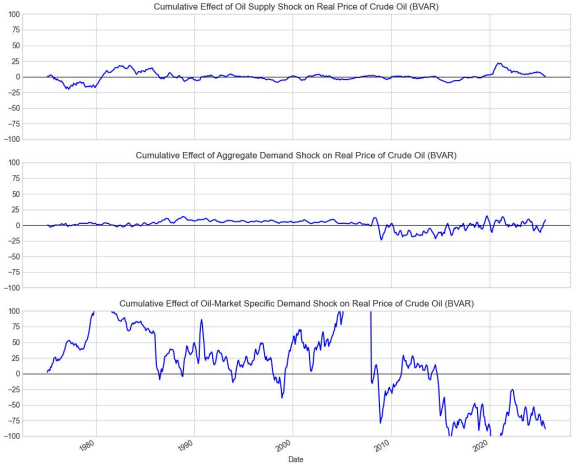
BSVAR



BSVAR (Out-of-sample)

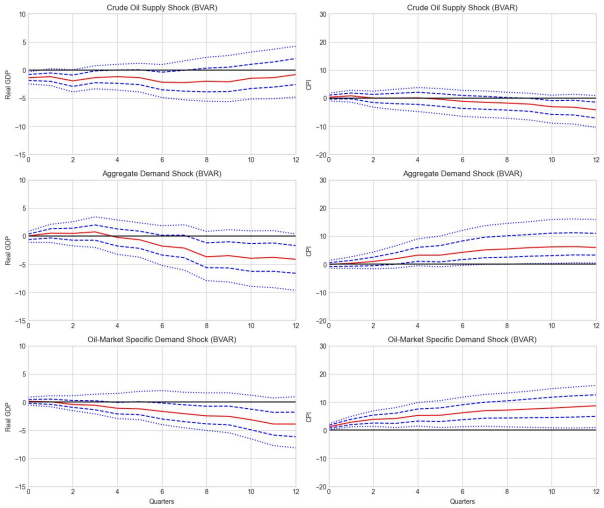


BSVAR (Full)

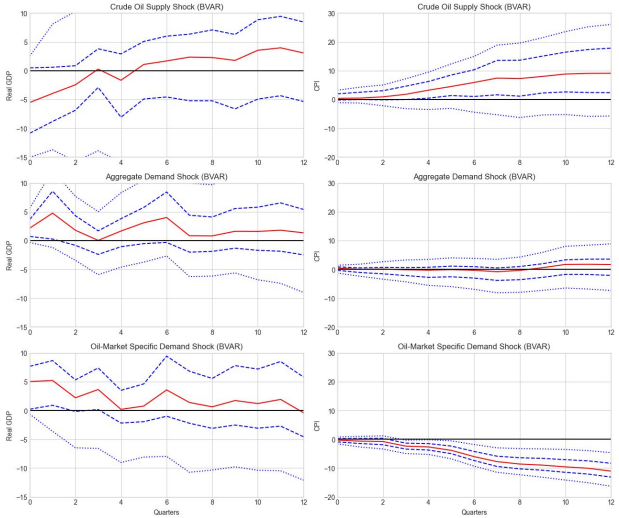


Empirical result - Effect on Economy

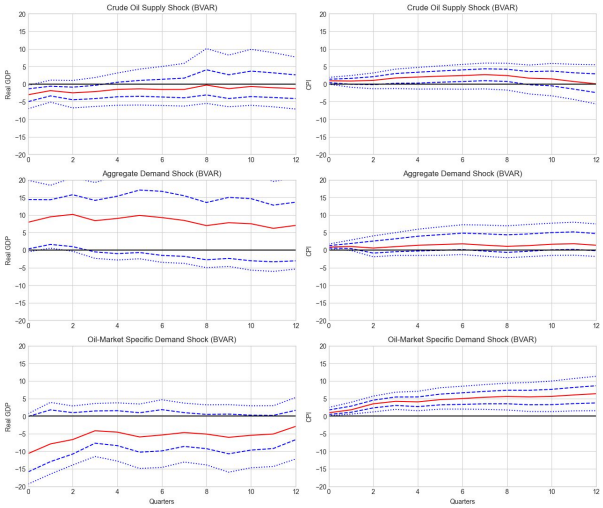
BSVAR



BSVAR (Out-of-sample)



BSVAR (Full)



Conclusion

- Kilian's original work showed that the **Demand is the main, consistent driver for the crude oil market**, **Supply is just a temporary shock**. The full sample seems to agree.
- But Out-of-sample study seems to disagree with his conclusion, it might be subjected to the **large market shock of 2020 April oil super-contango**.
- Also, shale production might have made production more flexible and less costly to adjust.
- Next step: Consider time-variant parameter model and modeling with identification of the 2020 oil price crisis.

Reference:

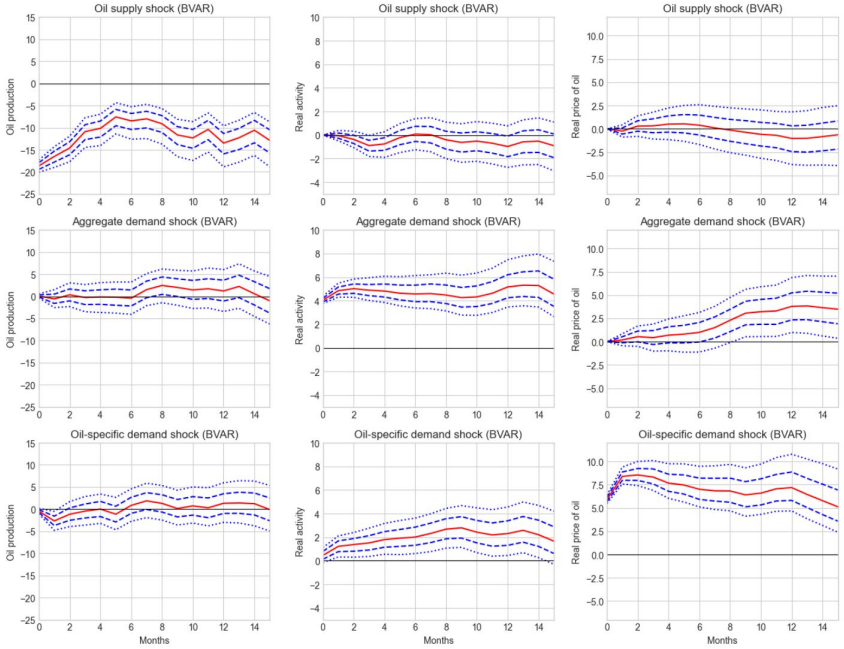
- Kilian, Lutz. 2009. "Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market." American Economic Review 99 (3): 1053–69.

Appendix - Yearly Events

Year	Event
1973	October War and Oil Embargo
1980	Iranian Revolution
1981	Iran-Iraq War
1990	Invasion of Kuwait
2001	OPEC Meeting
2002	Venezuela civil unrest
2003	Iraq War
2008–09	Financial Crisis
2010–12	Arab Spring
2014–16	Shale Boom
2015–16	Global Slowdown
2020	COVID-19 Pandemic
2022	Russia–Ukraine War

Appendix - TVP BSVAR

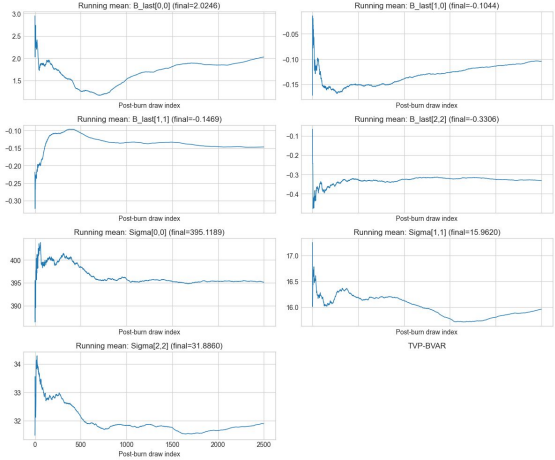
IRFs



$$Y(t) = \phi_0 + \phi_1 Y(t-1) + \phi_2 Y(t-2) + \ldots + \phi_2 4 Y(t-24) + \epsilon(t)$$

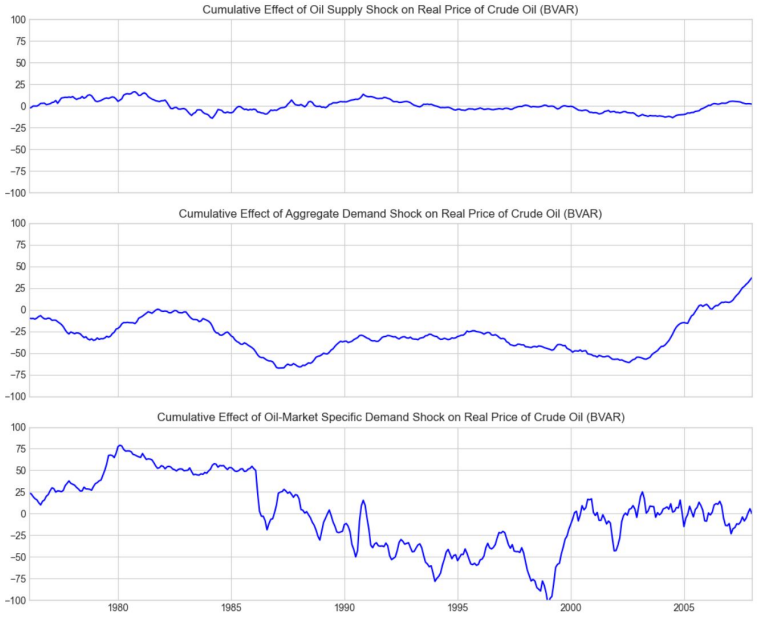
$$\phi_t = \phi_{t-1} + \eta_t : \eta_t \sim N(0, Q) \qquad \epsilon(t) \sim N(0, \Sigma), \Sigma \sim IW(\nu_0, S_0)$$

This model is significantly more computational expensive than any other model, we have to lower the lag term to 6 and sample draw to 3000 (500 burn-in) for a efficient computation (13 minute).

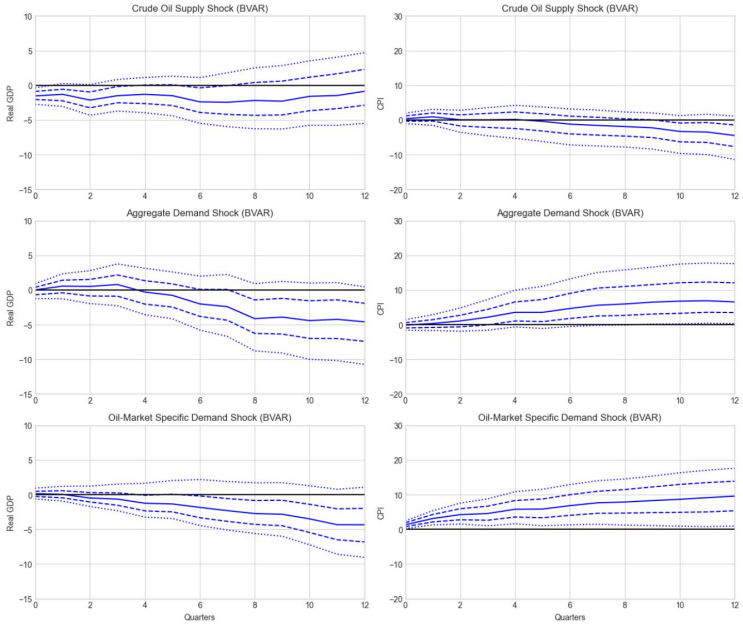


Appendix - TVP BSVAR

Cumulative Effect on Real Oil Price



Shocks on real GDP and CPI



Appendix - Lag Selection

Lag	AIC	Lag	AIC	Lag	AIC
1	16.426537	13	16.130171	25	16.301547
2	15.939740	14	16.138129	26	16.364076
3	15.958294	15	16.104418	27	16.410536
4	15.988829	16	16.063492	28	16.477417
5	15.981680	17	16.064639	29	16.460377
6	16.015324	18	16.110374	30	16.512006
7	16.068859	19	16.154497	31	16.550655
8	16.032343	20	16.116769	32	16.502422
9	16.016479	21	16.162367	33	16.497064
10	16.025294	22	16.229585	34	16.499710
11	16.027409	23	16.247498	35	16.476306
12	16.076206	24	16.278636	36	16.467682

We chose 24 lags as it is widely accepted in the literature (Kilian, 2009)

Opposing papers: Arampatzidis, 2026